



Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

Lab#01

Department of Computer Science (7th Semester)

Course: DevOps

Name:

Zahra Bashir

SUBMITTED TO:

Sir Waqas Saleem

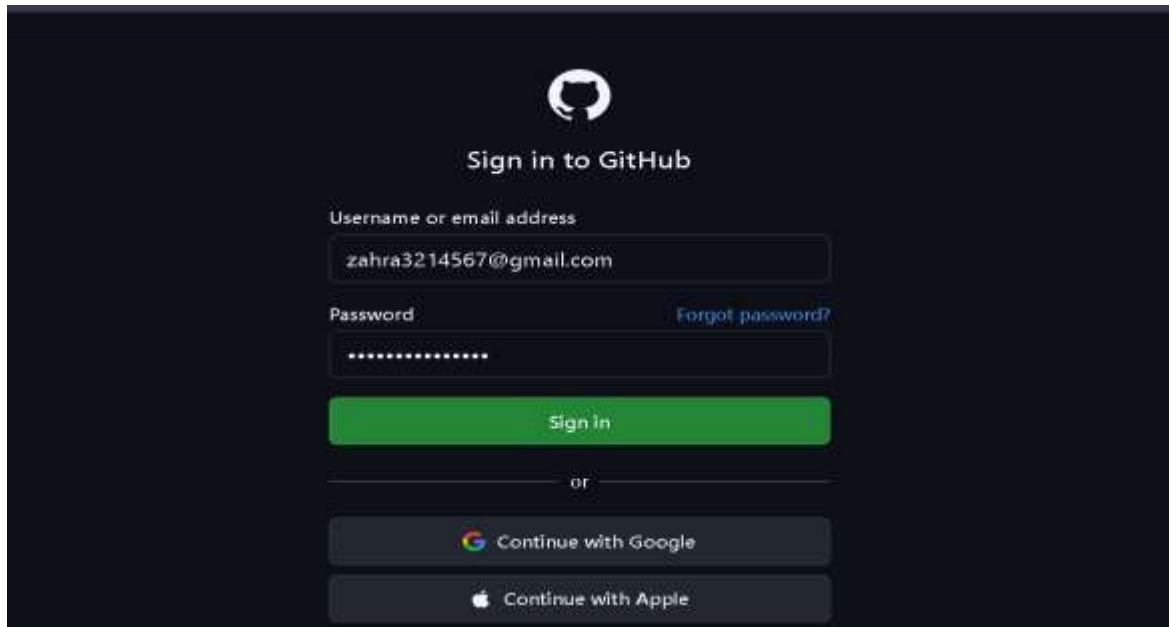
Registration Number

23-BCS-092(Section-A)

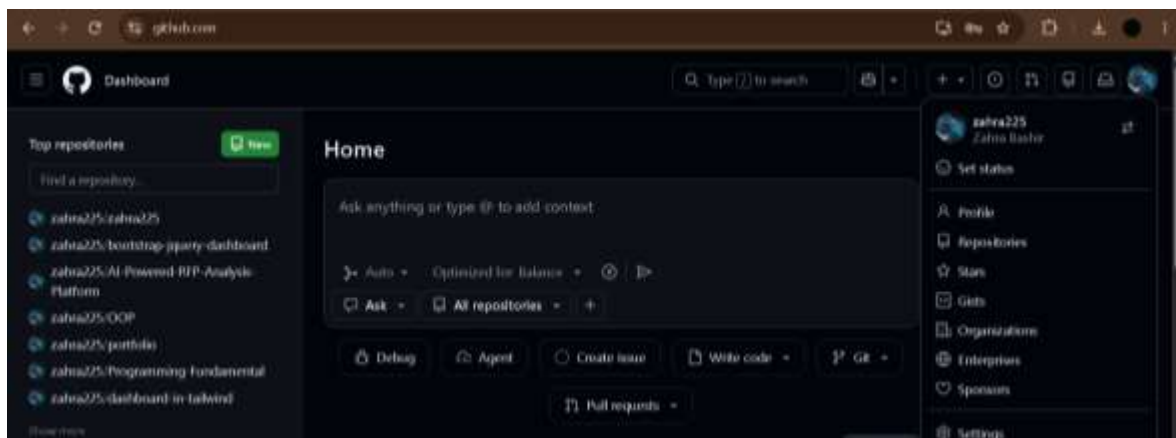
GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Task 1: Create and Configure Your GitHub Account

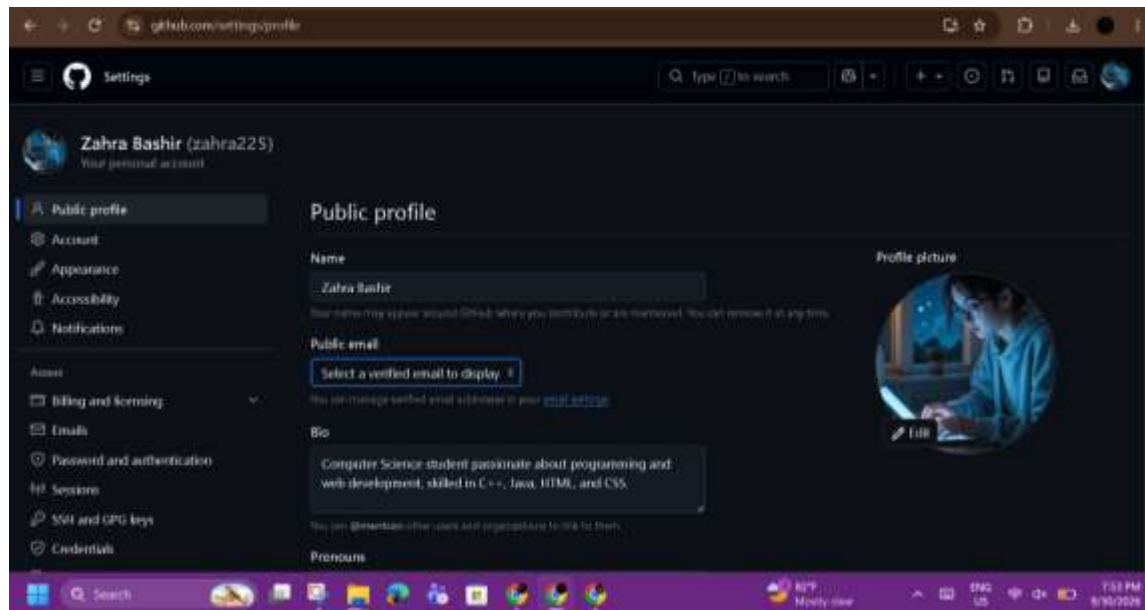
1. Open [GitHub Sign Up](#) and create your personal GitHub account.



2. Choose a professional GitHub username related to your actual name.

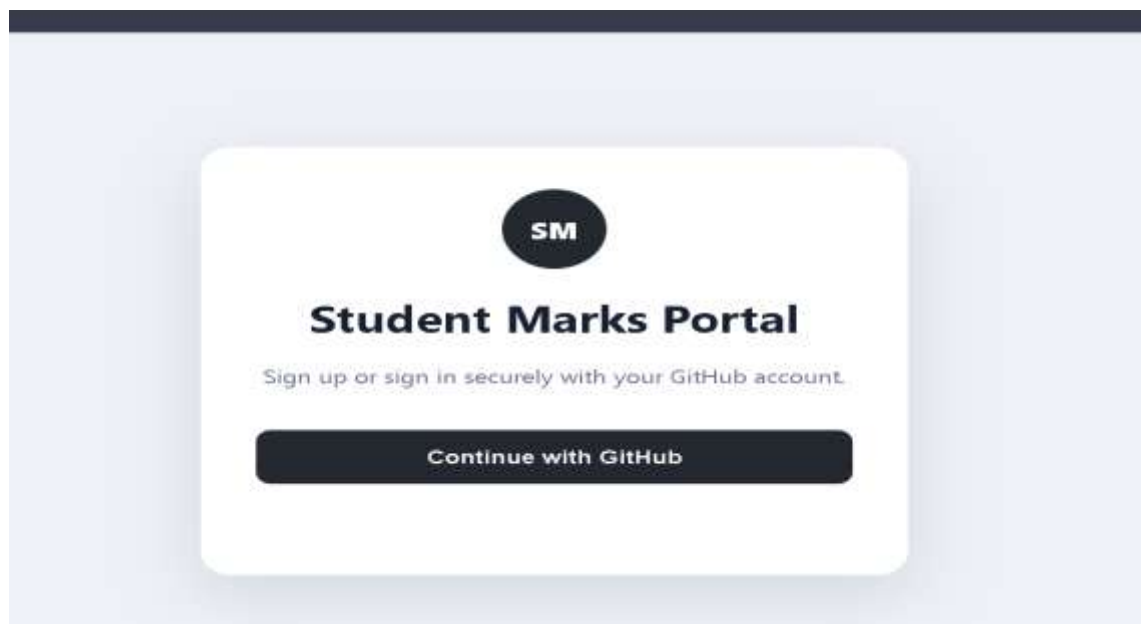


3. Open GitHub > Settings > Public profile.

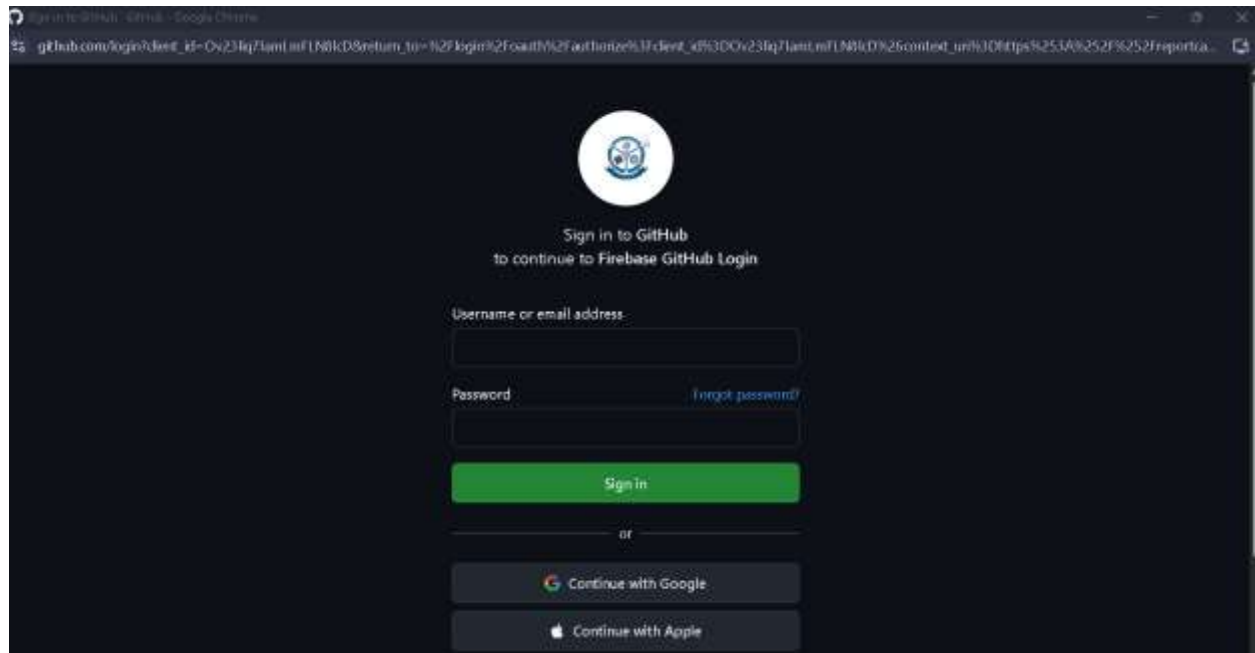


Task 2: Register in the Student Marks Portal

1. Open the [Student Marks Portal](#).



2. Select **Continue with GitHub** and sign in using the GitHub account created in Task 1.



3. Enter your actual registration number using the format provided by your instructor.

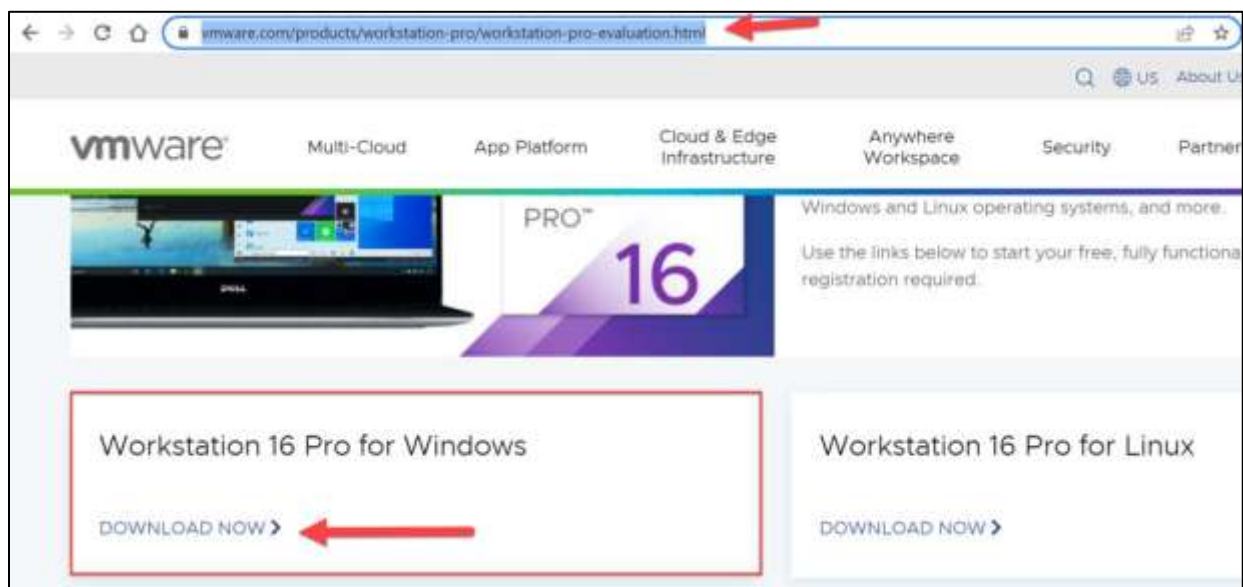
A screenshot of a web browser showing a form titled "Complete registration". The form has a light blue background. At the top, it says "Enter your registration number, course and section. The enrollment requires approval." Below this, there are three input fields: "Registration number" with the value "2023-85CS-92", "Course" with the value "CS-649 -- Cloud DevOps", and "Section" with the value "Section A". Below these fields are two buttons: "Submit enrollment" and "Sign out".



Task 3: Download VMware Workstation Pro and Ubuntu Server

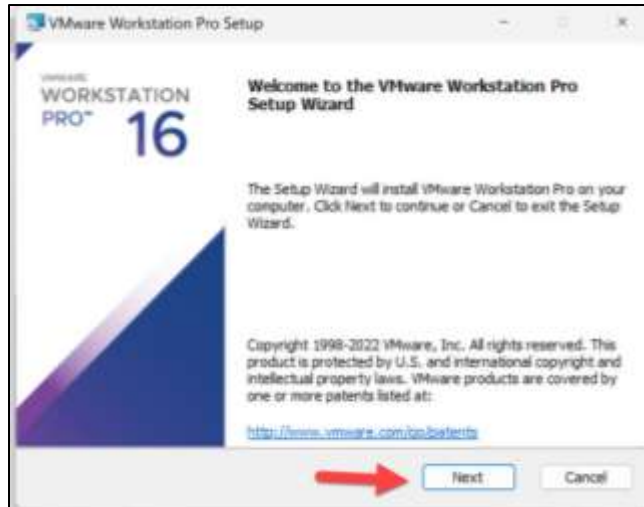
Installation of VMware and Ubuntu:

Step 1. Download VMware Workstation Pro for Windows



Step 3. Install VMware Workstation Pro on Windows 11

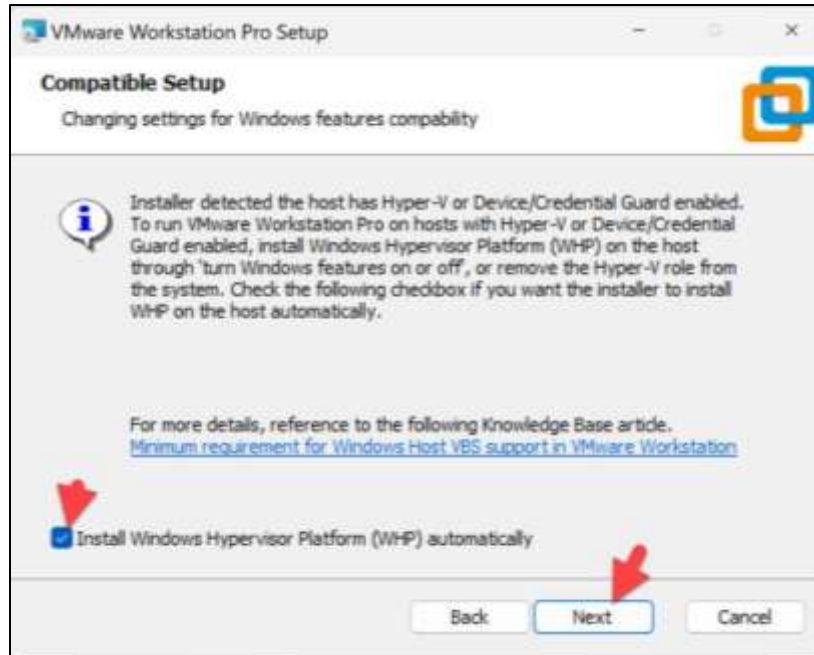
- To begin, locate and run the installer after it has been downloaded, and click “Next” to begin the installation.



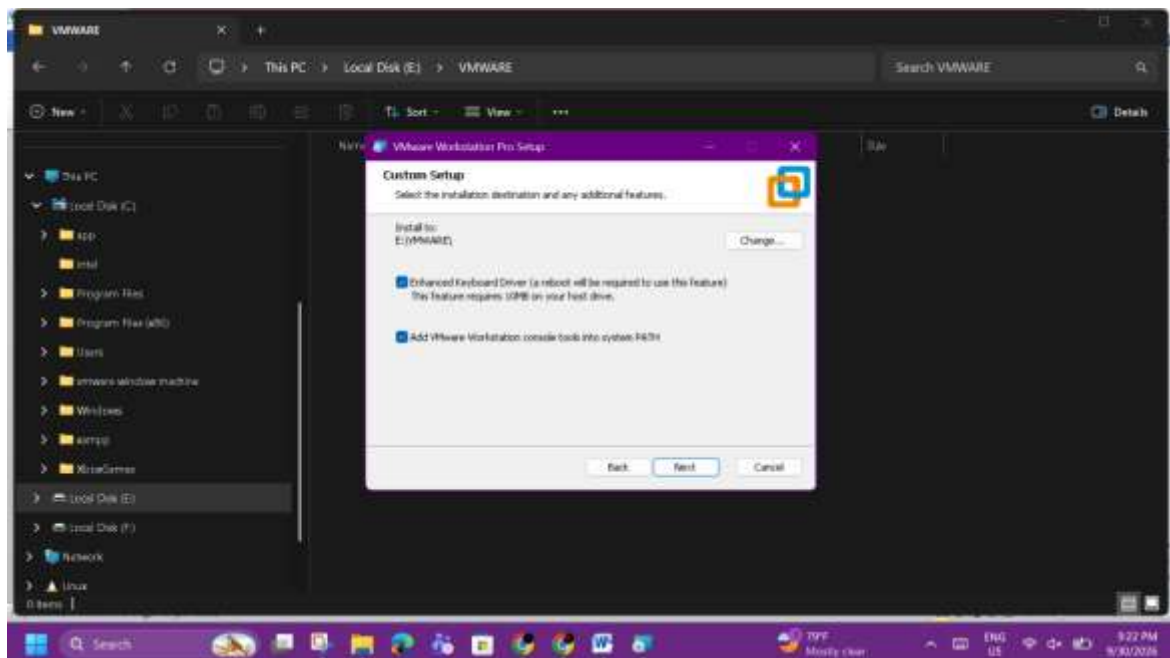
- Following that, you must agree to the License Agreement conditions and proceed to the next screen by clicking “Next.”



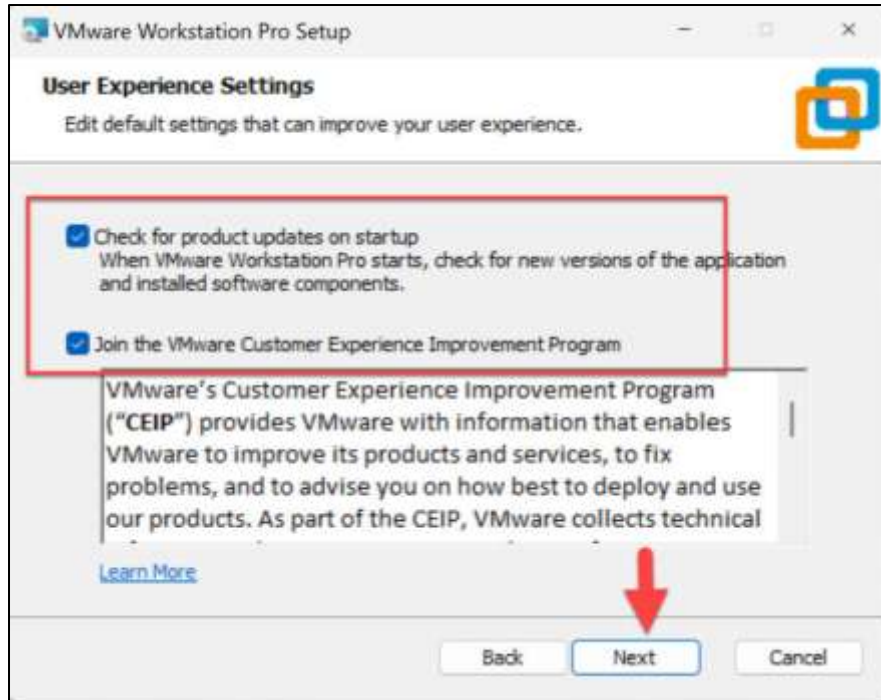
- Check the box for “Install Windows Hypervisor Platform (WHP) automatically,” and click “Next.”



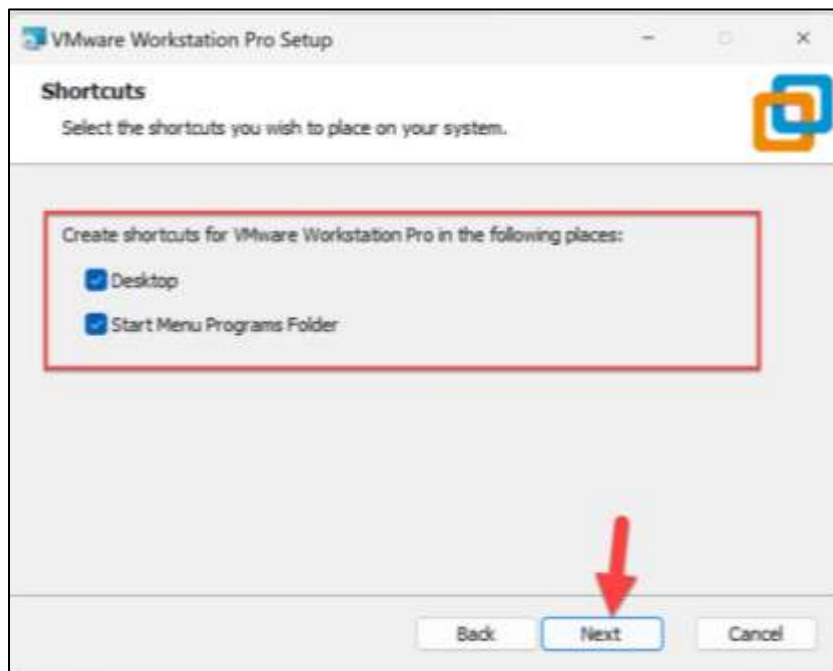
- Tick the box next to “**Enhanced Keyboard Driver**” to integrate the keyboard effectively. In addition, select the option to add other console tools to the system PATH. Click “**Next**” to continue.



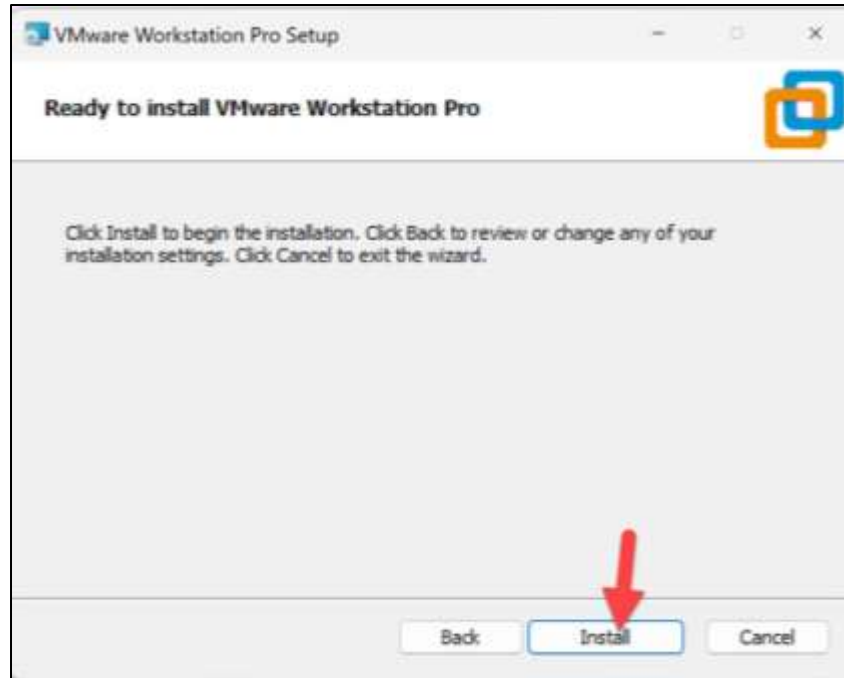
- In User Experience Settings, select your preferred settings and click “**Next**” to proceed



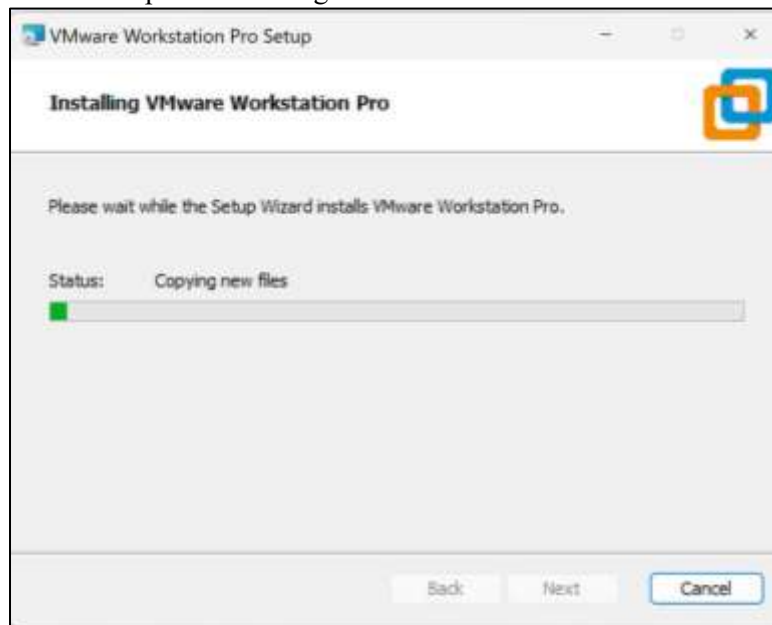
- You can accept or change the default installation location on the following page. When done, click “Next.”



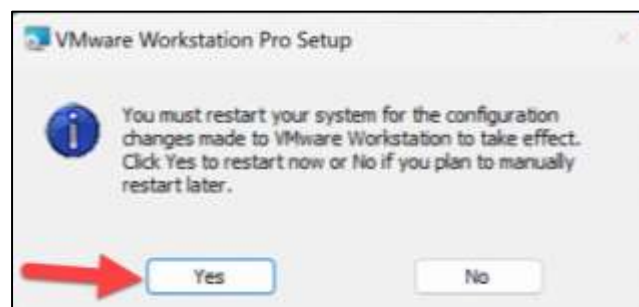
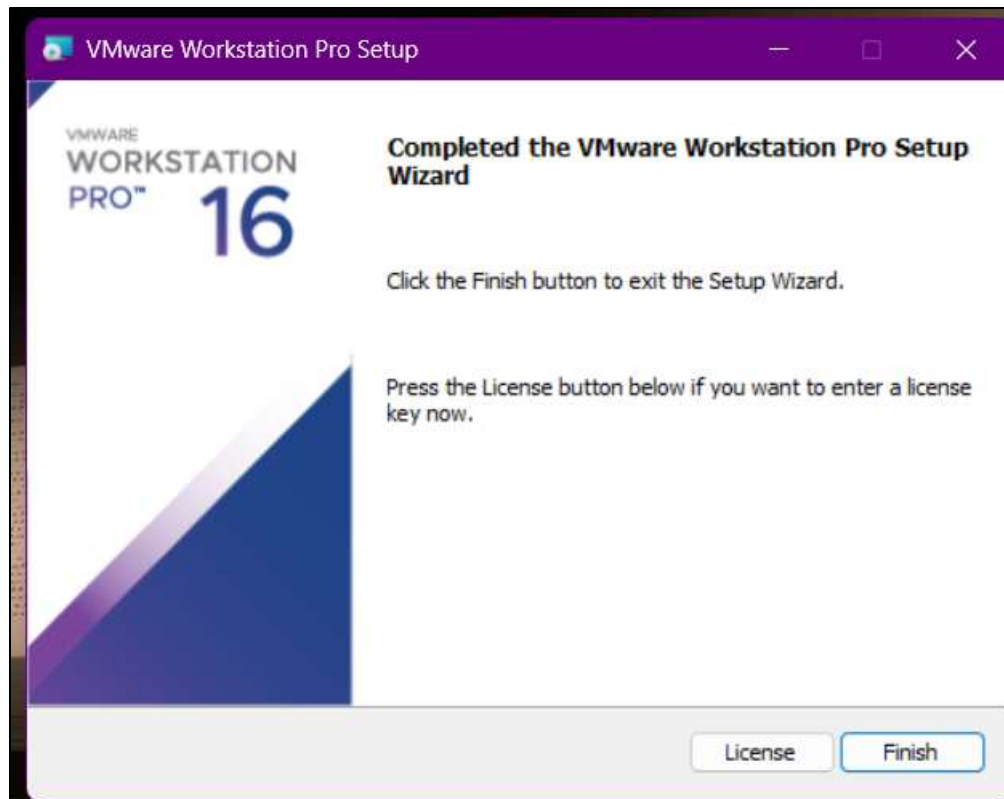
- Finally, click the “**Install**” button on the screen to begin the installation.



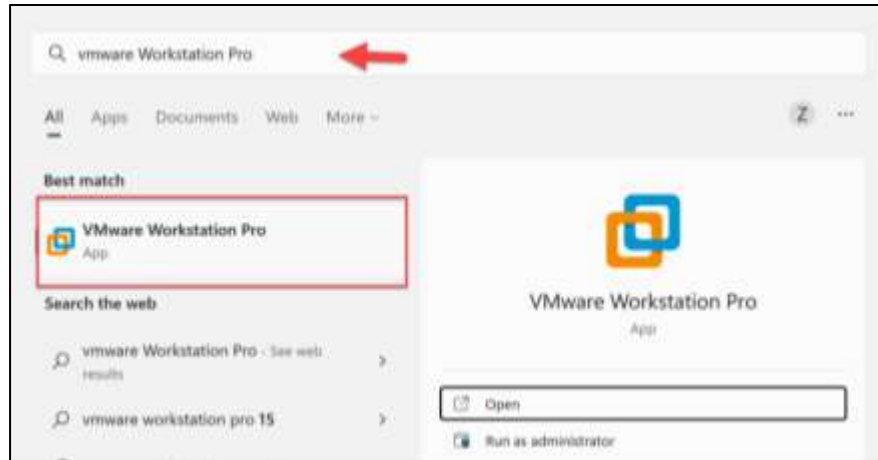
- The VMware Workstation Pro installation process should take only a few minutes. Click the “**Finish**” button to complete the configuration.



- Next, you need to restart your system for the changes to take effect. For that, click “**Yes.**”



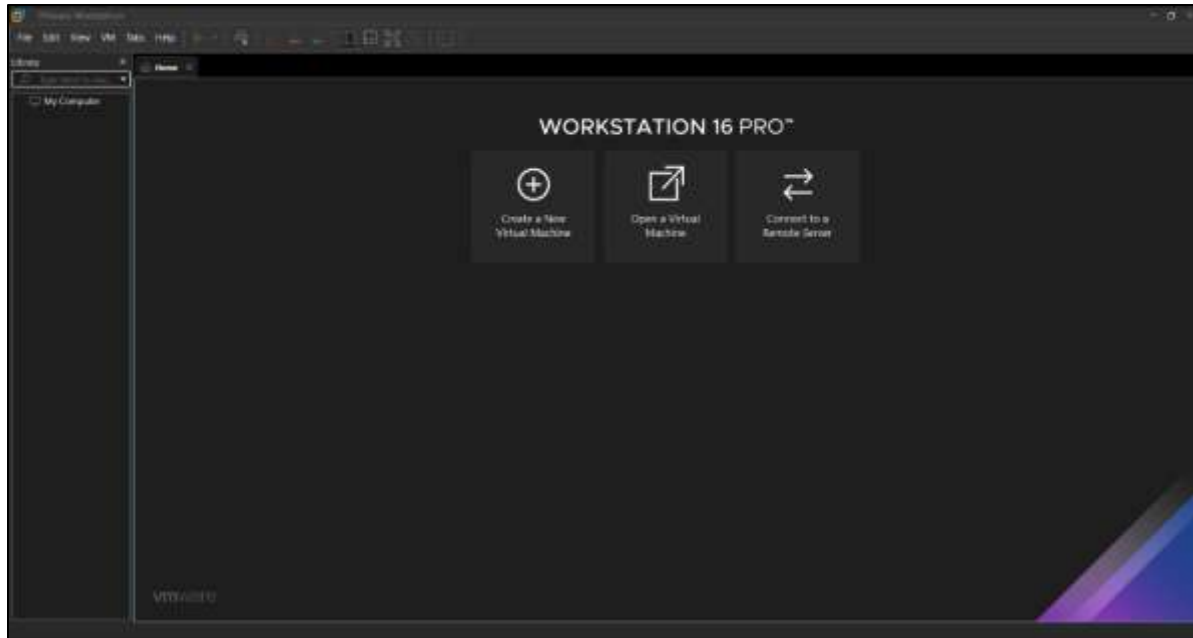
- After your computer has restarted, open VMware Workstation Pro by searching for it in the Start Menu.



- Finally, click “**Finish**,” and the program will start automatically.

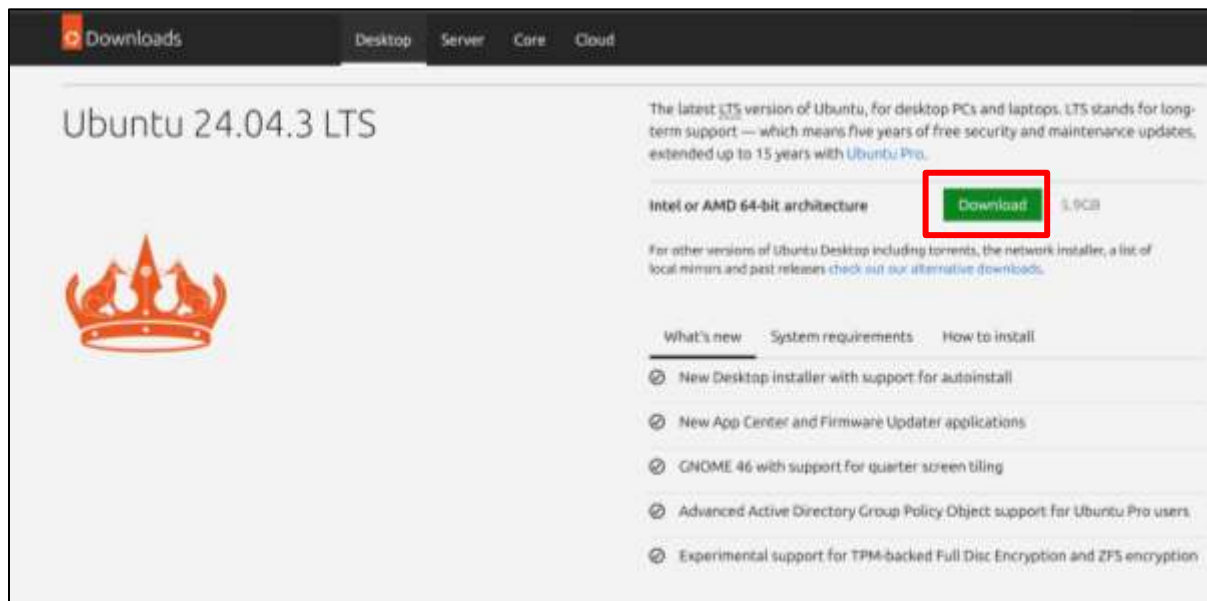


- That's it! VMware Workstation Pro is installed on your PC. Now, you can create a new virtual machine and try another operating system along with Windows.

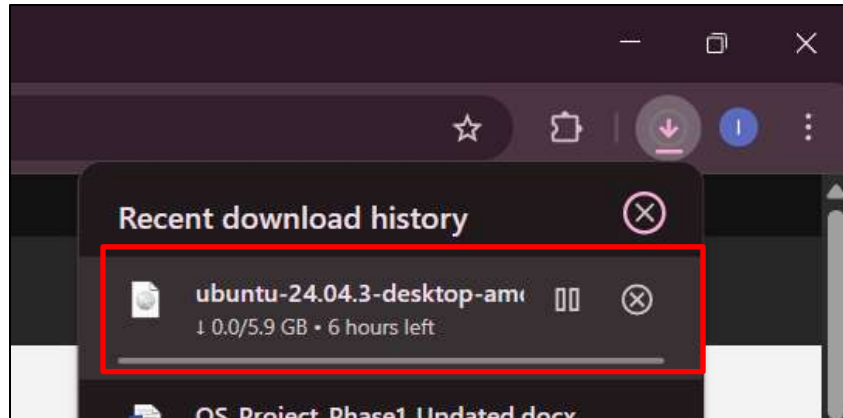


- The Ubuntu Desktop ISO file was downloaded from the official Ubuntu site.

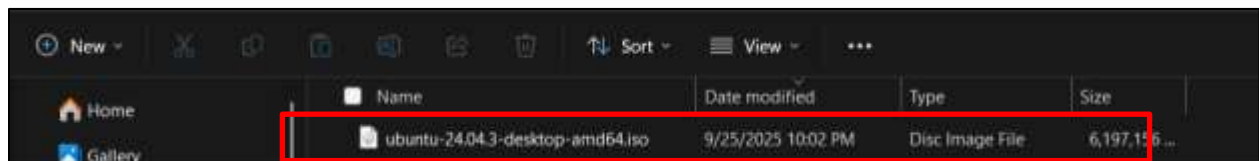
Step 1. Visit the site and download the Ubuntu ISO file.



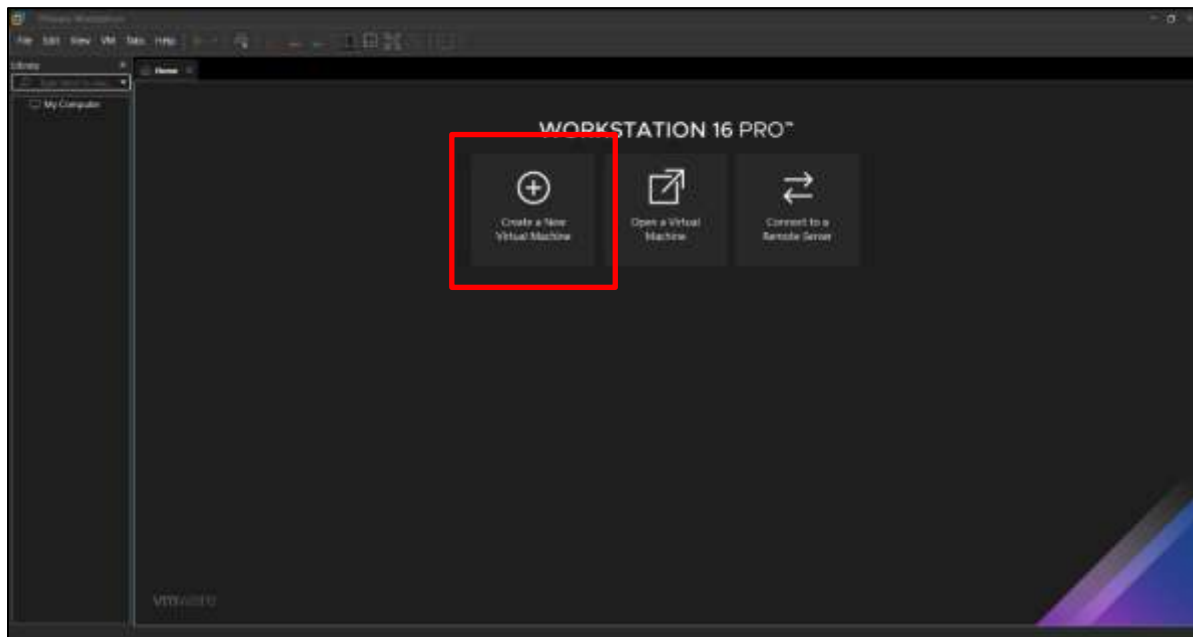
Step 2. The file will start to download.

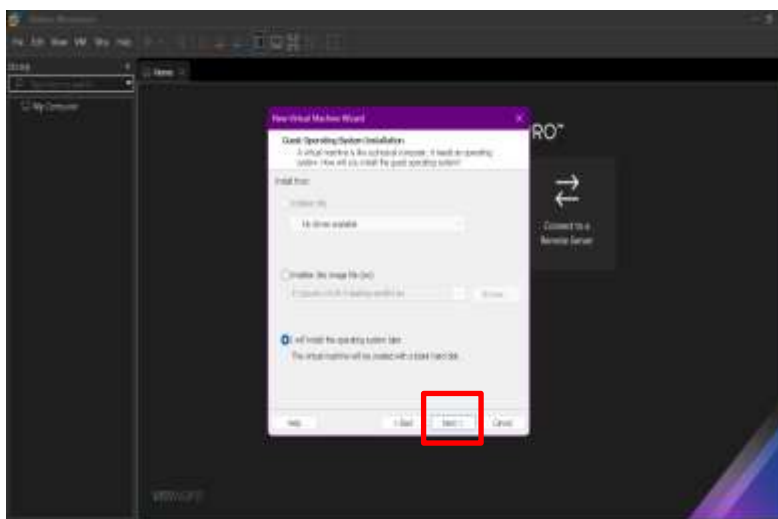
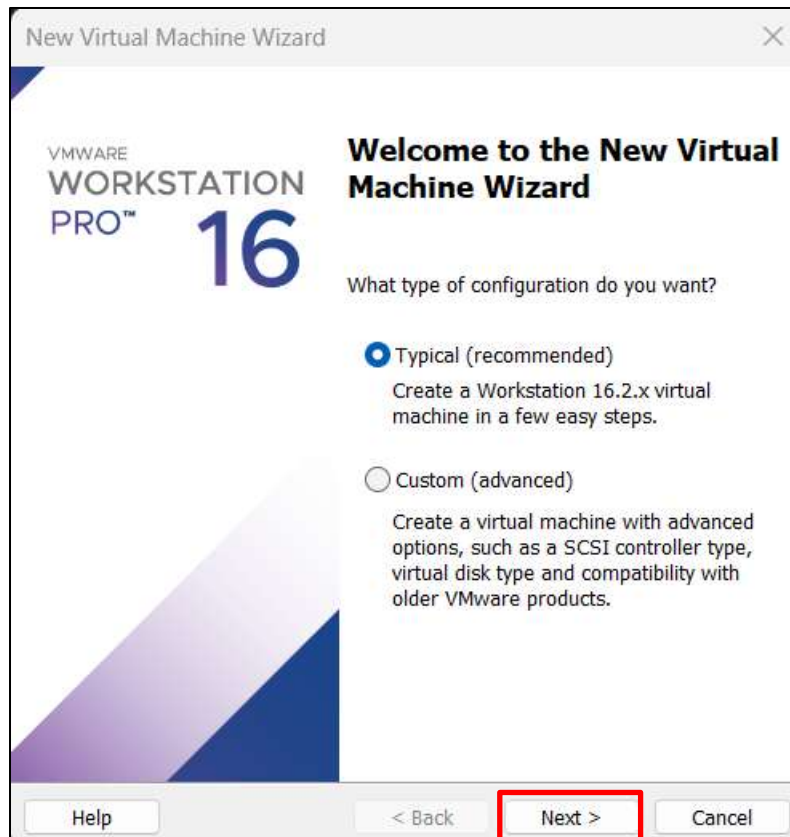


Step 3. After the download finishes, the ISO file will appear in the downloads folder.



Task 4: Create the Ubuntu Server Virtual Machine





New Virtual Machine Wizard

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

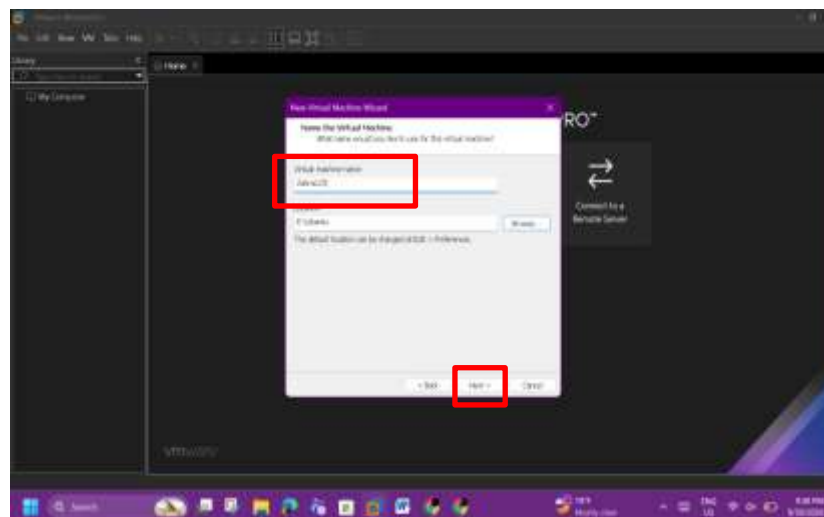
☐ VMware ESX

☐ Other

Version

Ubuntu

Help < Back Next > Cancel



New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu: 20 GB

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

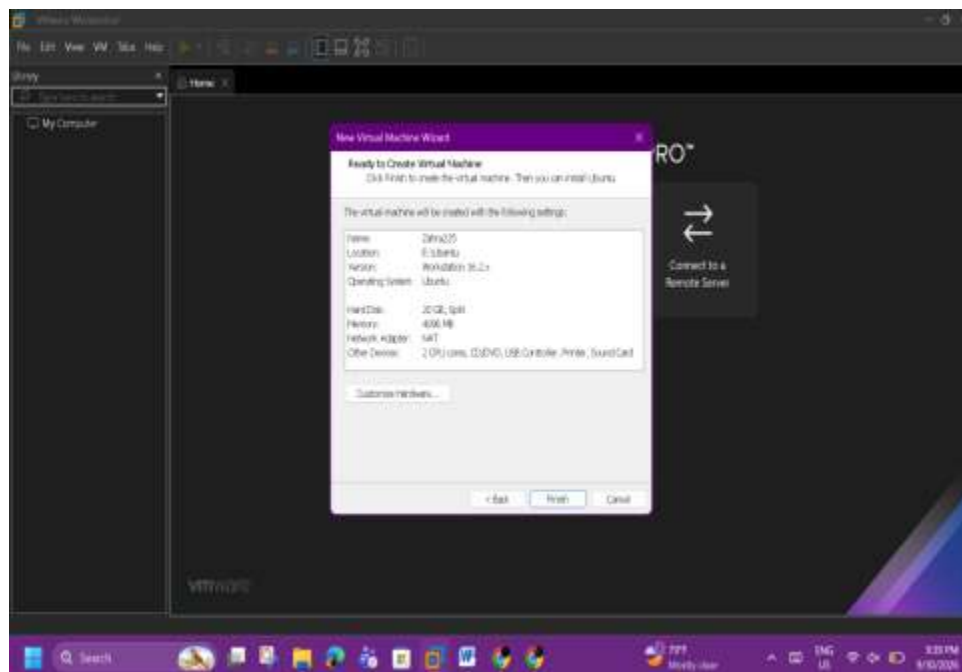
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

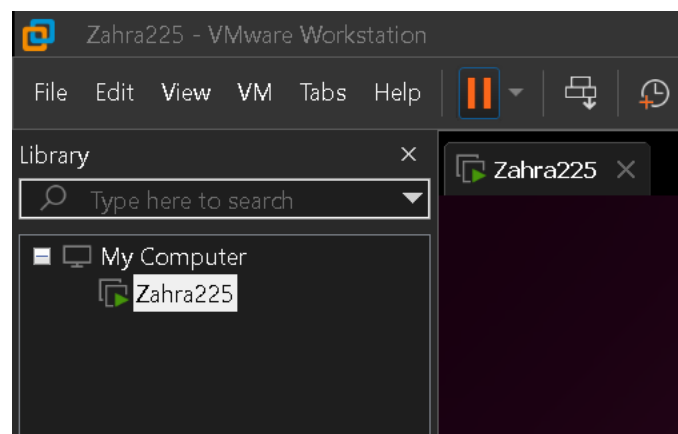
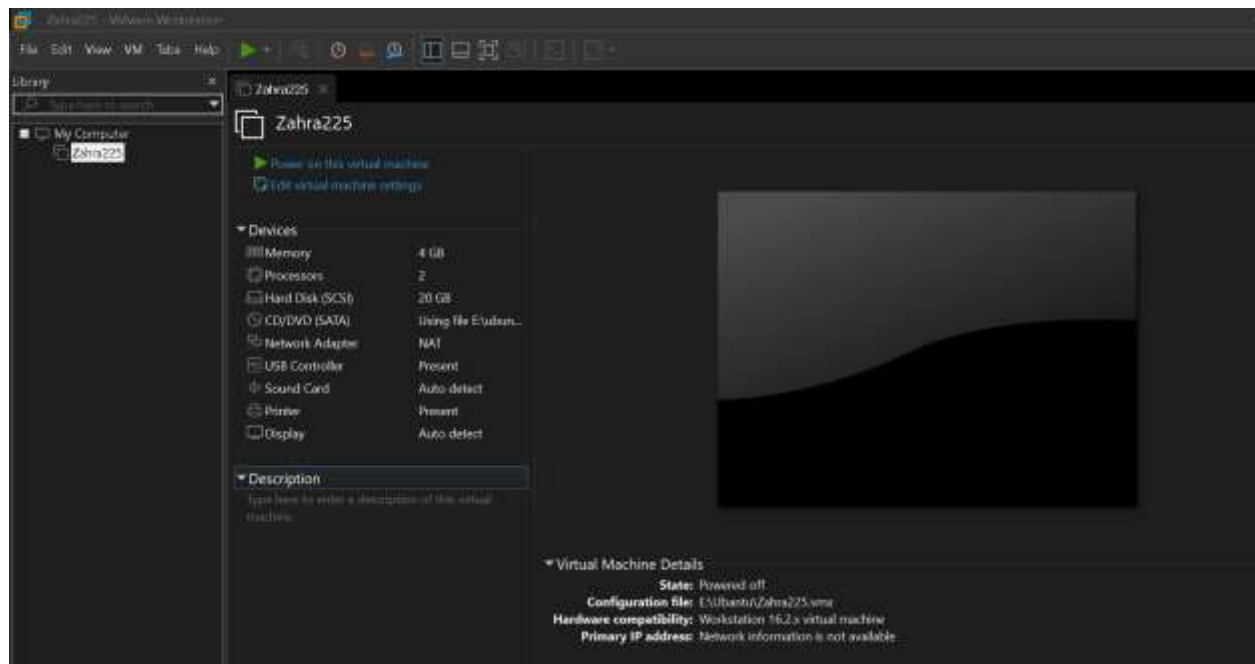
Help

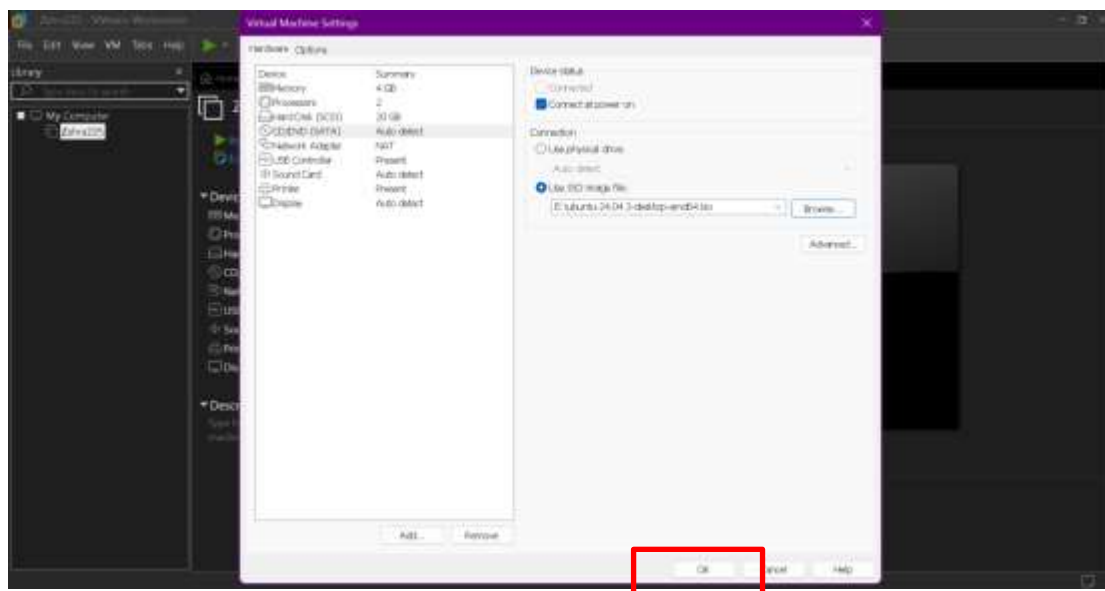
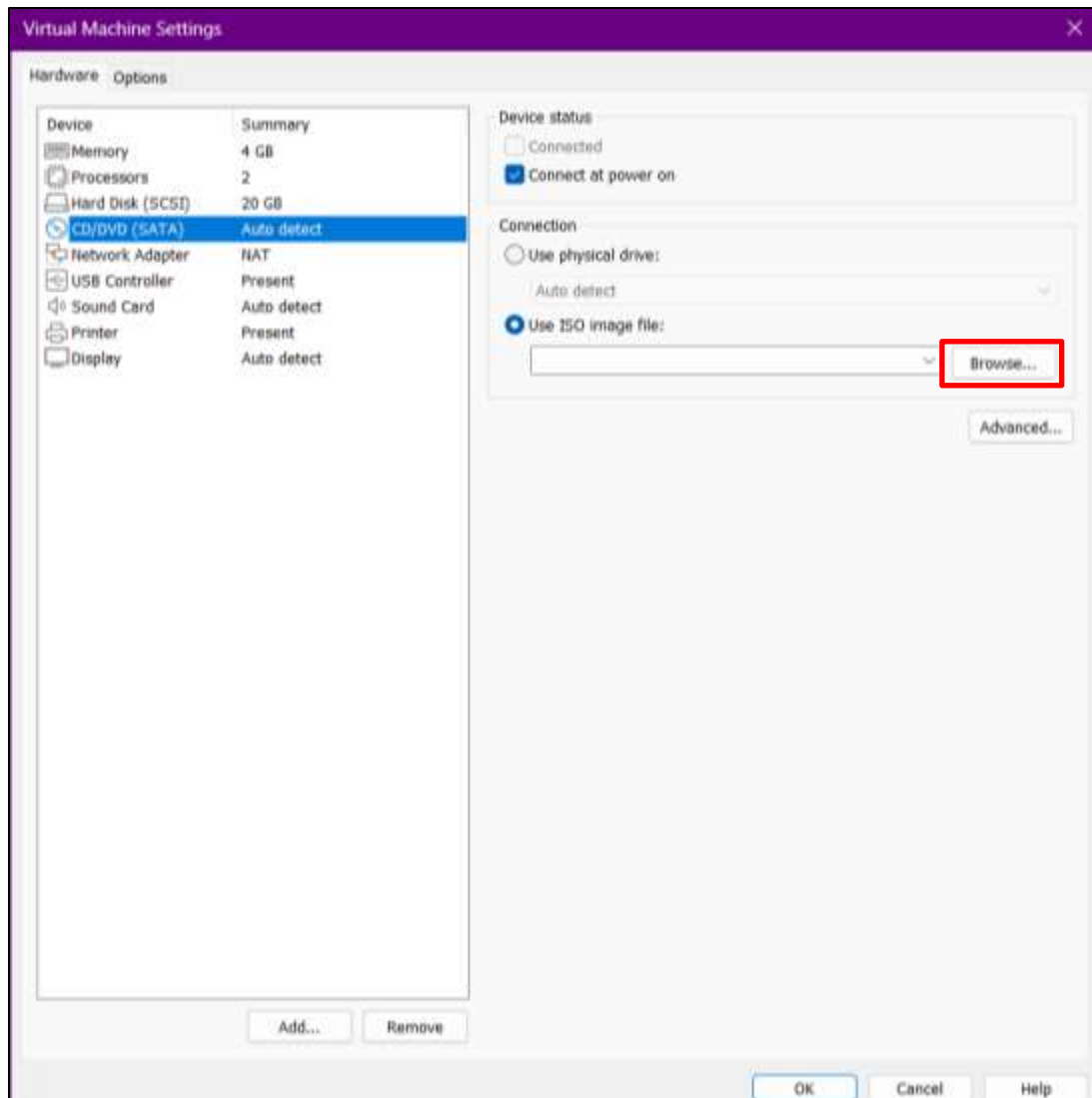
< Back

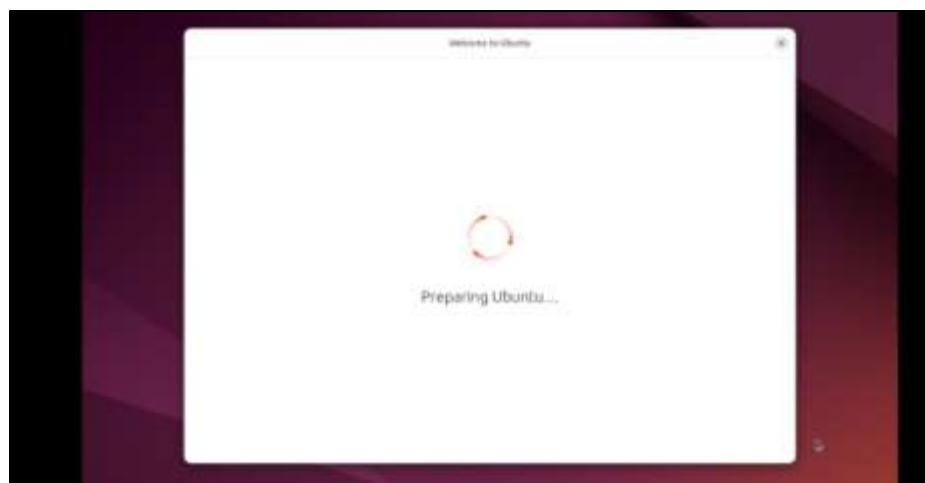
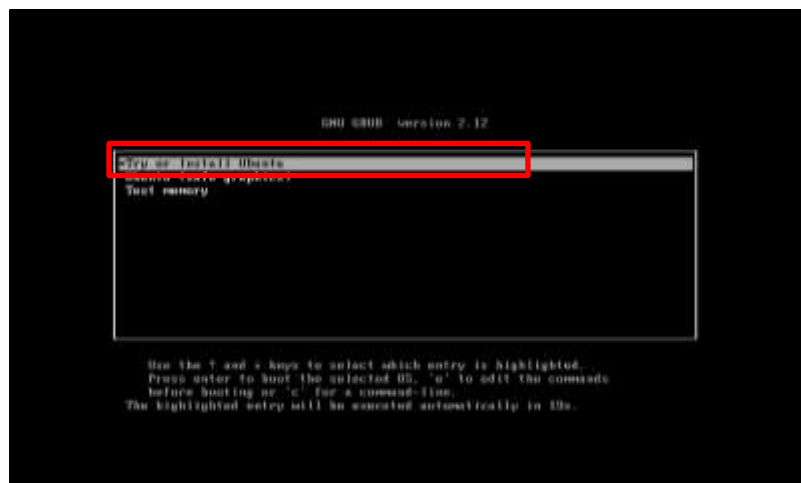
Next >

Cancel

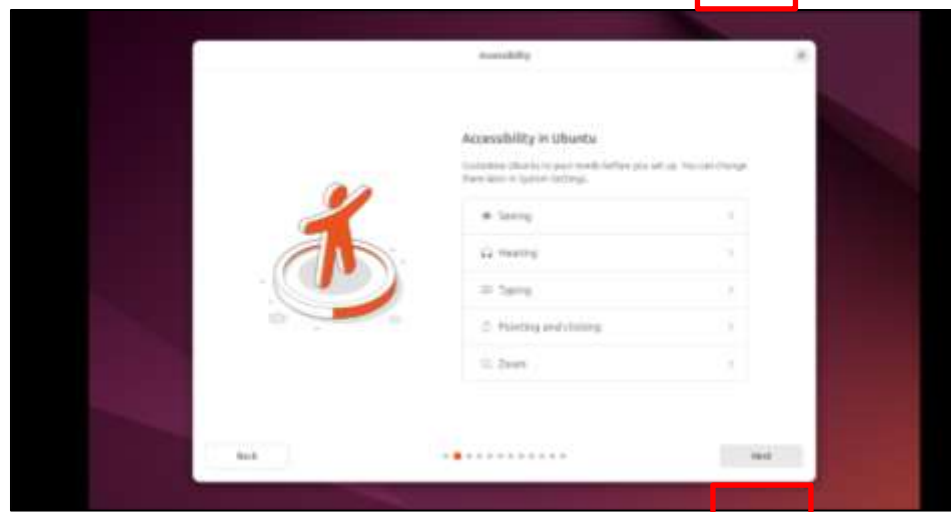




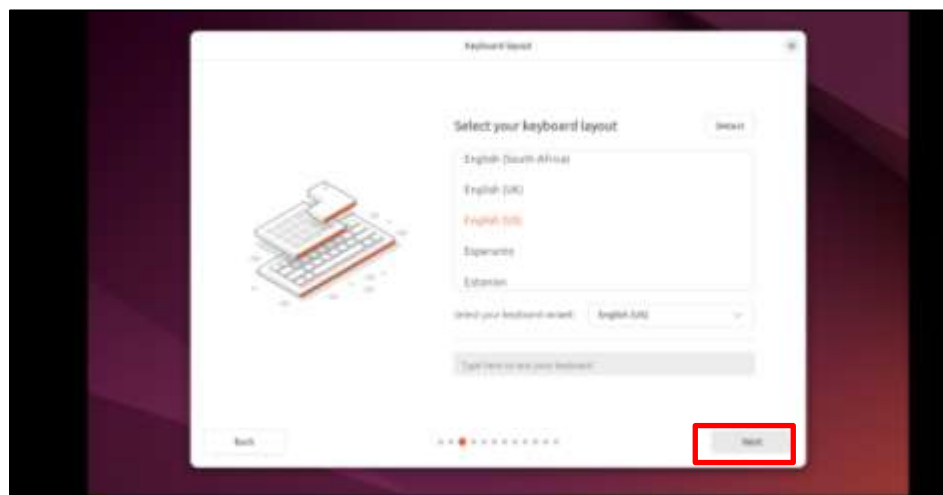


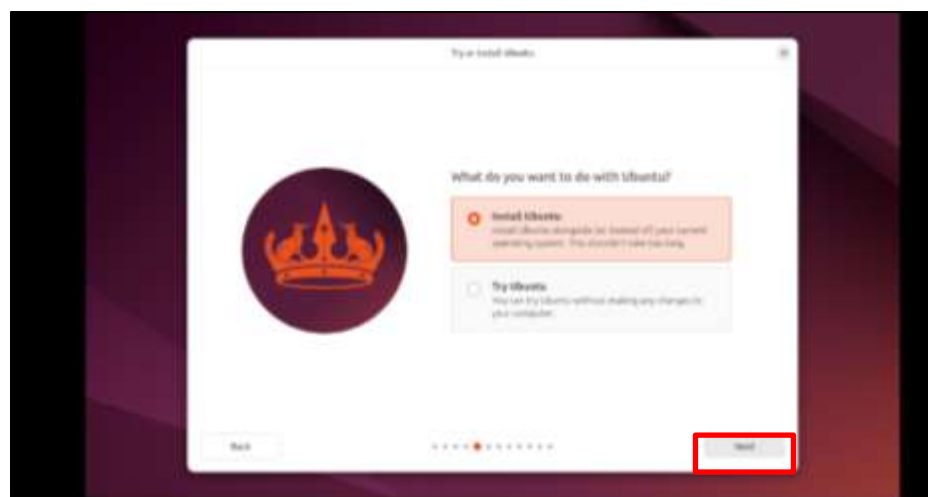
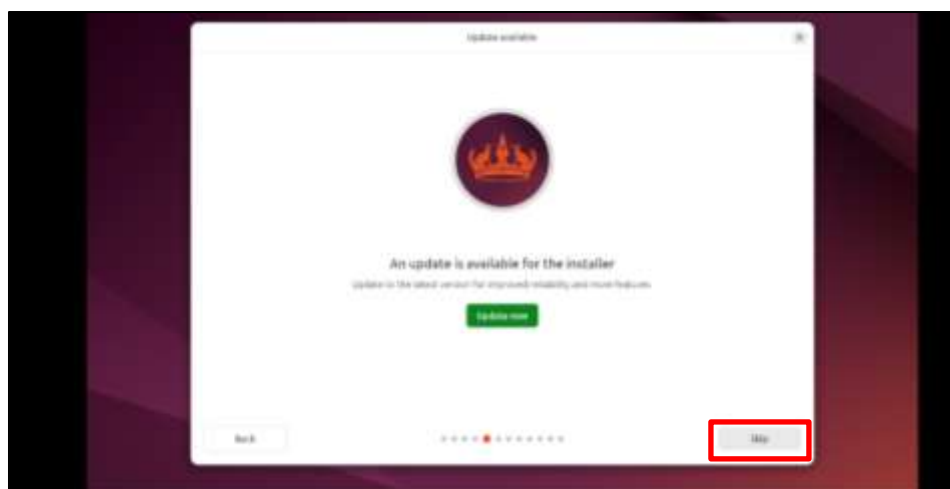
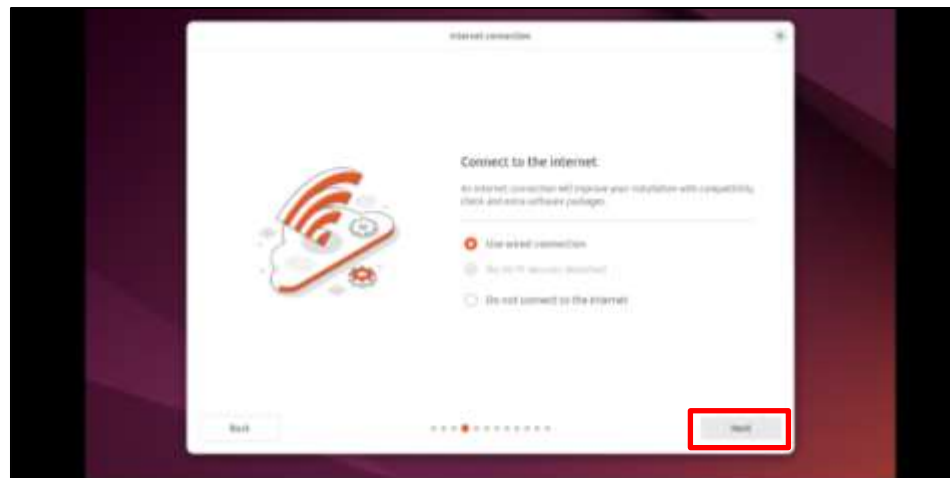


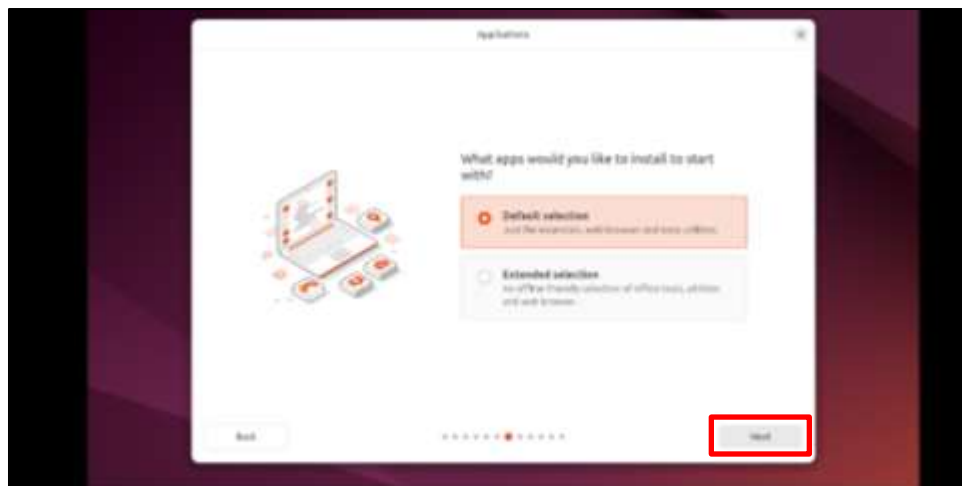
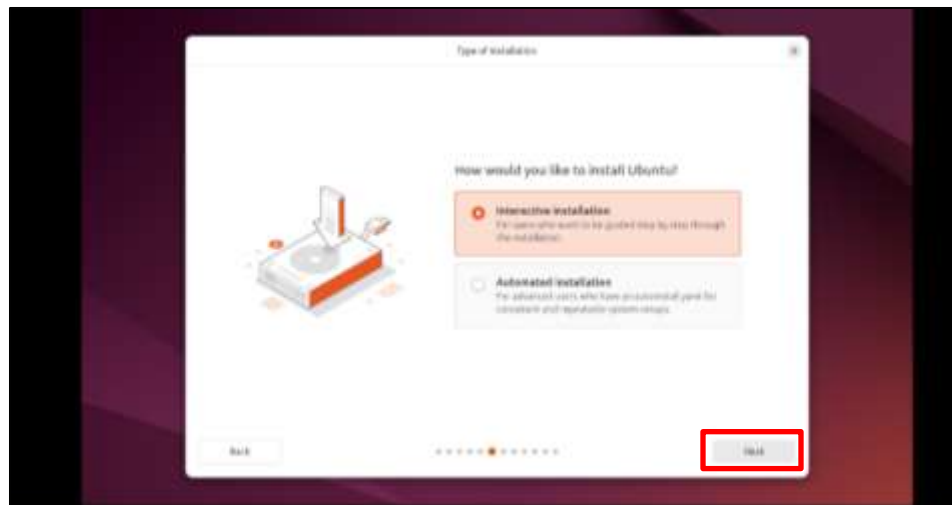
Task 5: Choose the Installation Language

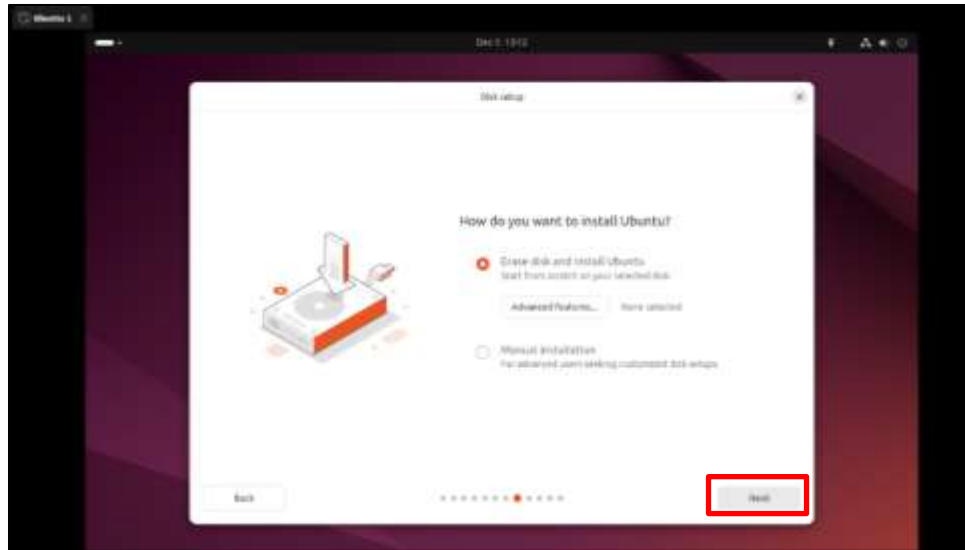


Task 6: Configure the Keyboard Layout

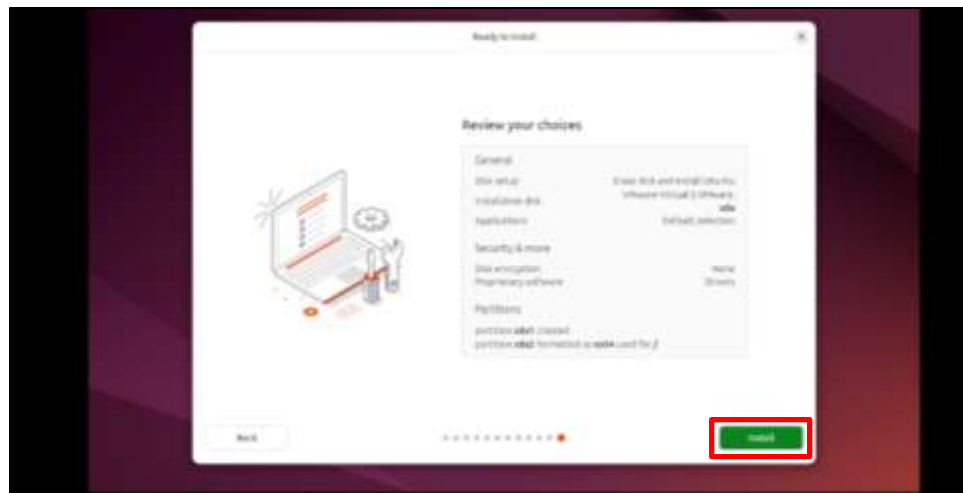


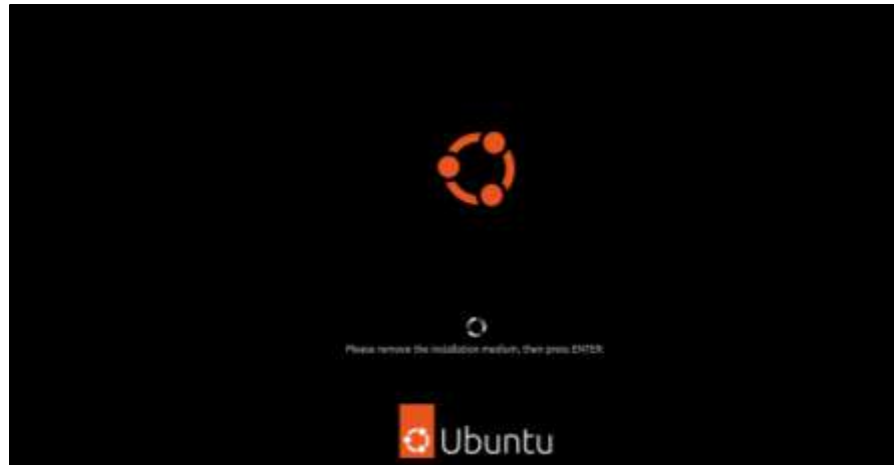




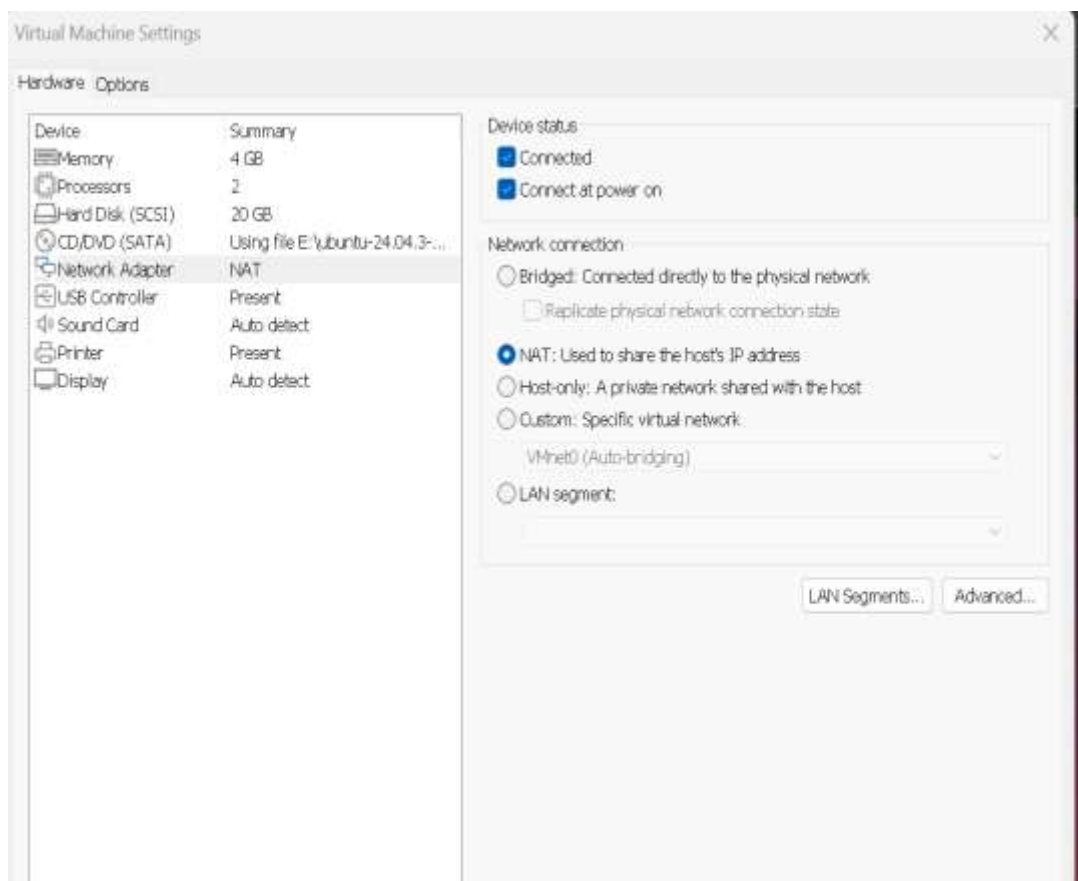


Task 7: Choose the Ubuntu Server Installation Type





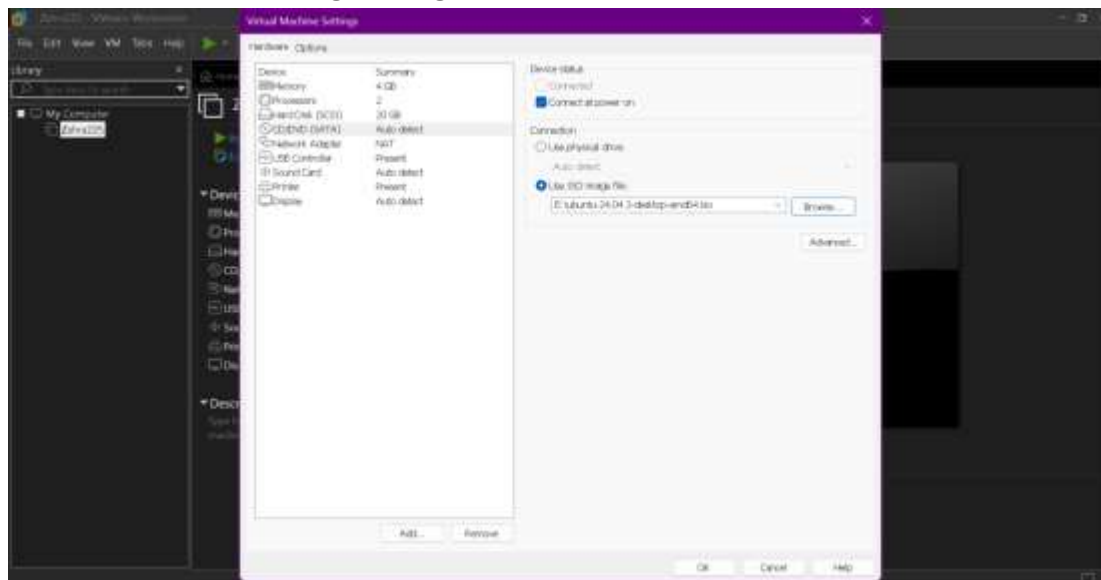
Task 8: Configure the Network



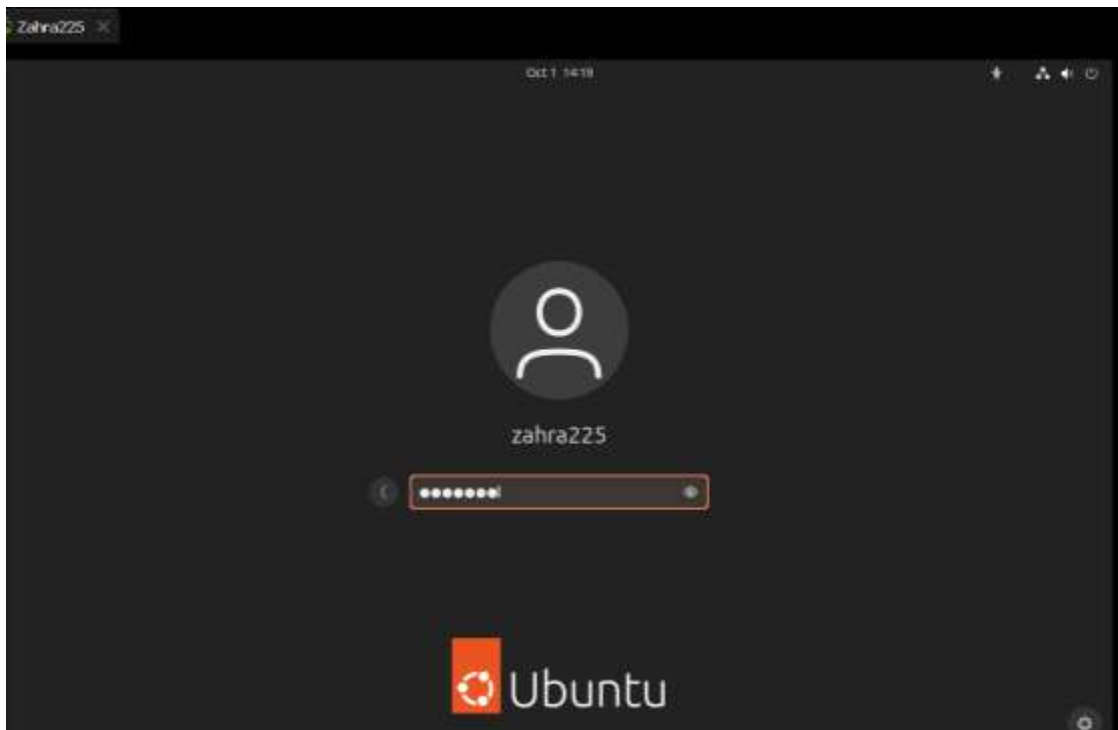
Task 9, 10, 11: Configure Storage. Select the Installation Disk. Review the Partitions

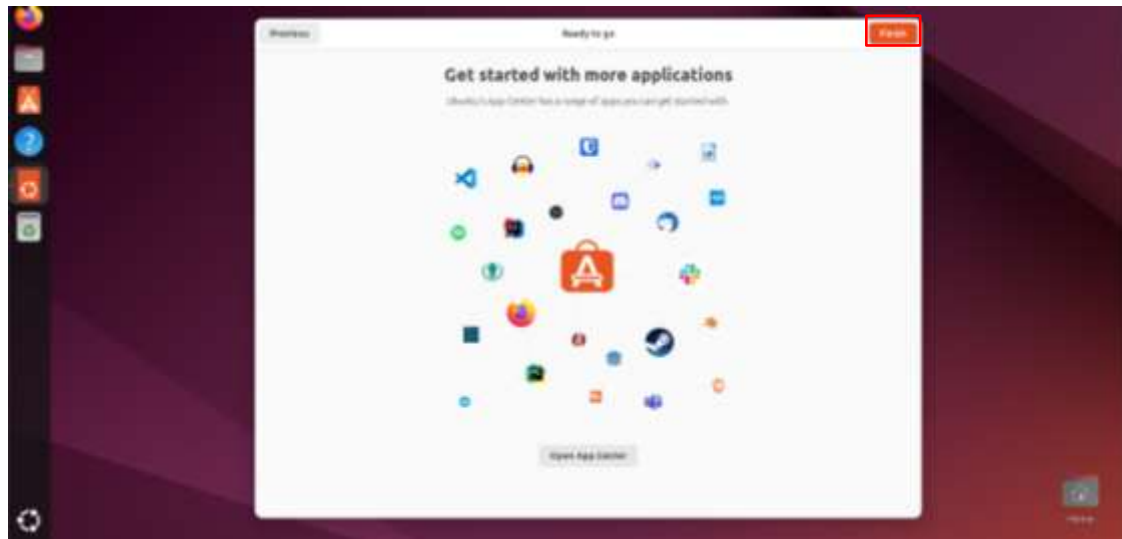
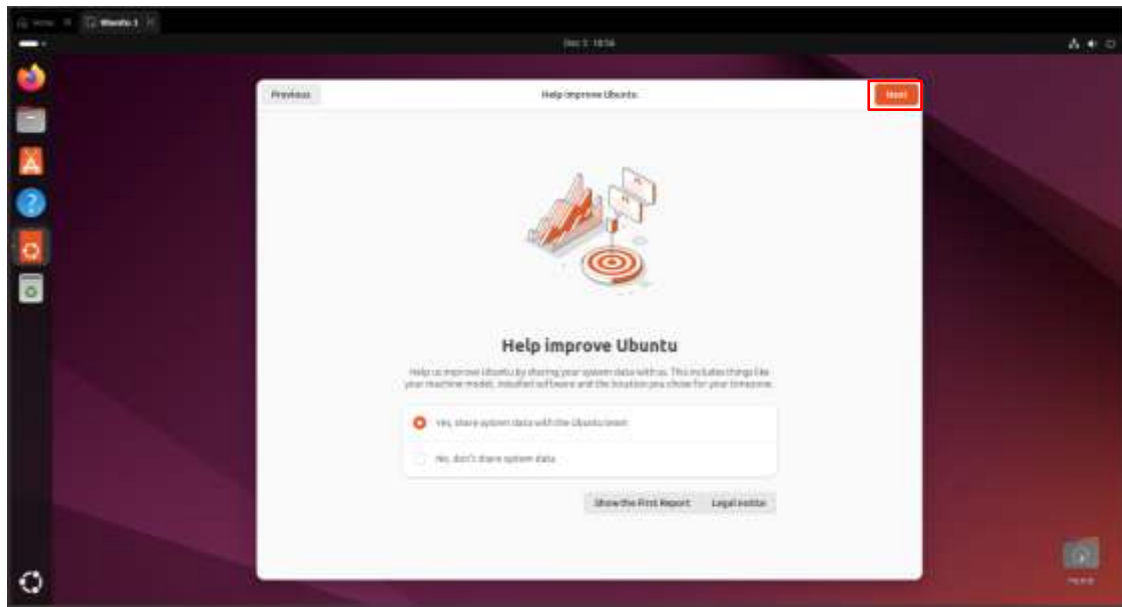


Task 12: Confirm the Storage Changes



Task 13: Configure the Ubuntu Profile







Task 16: Find the Ubuntu Server IP Address

A screenshot of a terminal window titled 'Zahra225'. The terminal shows the output of the 'ip addr' command. The output lists network interfaces: 'lo' (loopback) and 'ens33' (ethernet). The 'lo' interface has an IP address of 127.0.0.1. The 'ens33' interface has an IP address of 192.168.255.120. The terminal text is as follows:

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN  
    link/ether 08:0c:29:1d:81:3e brd ff:ff:ff:ff:ff:ff  
    altname eno2s1  
    inet 192.168.255.120/24 brd 192.168.255.255 scope global dynamic noprefixrou  
to ens33  
        valid_lft 1690sec preferred_lft 1690sec  
    inet6 fe80::20c:29ff:fe1d:813e/64 scope link  
        valid_lft forever preferred_lft forever  
zahra225@zahra225-Virtual-Platform: $
```

Task 17: Connect to Ubuntu Server from Windows

```
PS C:\WINDOWS\System32> ssh zahra225@192.168.255.128
The authenticity of host '192.168.255.128 (192.168.255.128)' can't be established.
ED25519 key fingerprint is SHA256:X8a5fp5LKbPdzy1B7a3GZvCL7nBGKyRWMup0yiLxVaM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.255.128' (ED25519) to the list of known hosts.
zahra225@192.168.255.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-27-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

507 updates can be applied immediately.
364 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

zahra225@zahra225-VMware-Virtual-Platform:~$
```